ARISHA AFZAL

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EDUCATION

Nadirshaw Eduljee Dinshaw (NED) University, Karachi, Pakistan — Oct. 2018 - Oct. 2022

Bachelor of Engineering in Electronic Engineering

Coursework: Probability and Statistics; Programming Languages; Data Structure and Algorithms; Linear Algebra & Geometry; Microprocessor Programming & Interfacing; Business Communication and Ethics

TECHNICAL SKILLS

Python (Pandas, Numpy, Matplotlib), SQL, Microsoft Power BI, Microsoft Excel (pivot tables, formulas, charts), C, MATLAB, VS Code, Git/Github, Claude Code, Prompt Engineering.

PROFESSIONAL EXPERIENCE

K-Electric | *Data Analysis, Data Visualisation, Reporting* — Karachi, Pakistan

Assistant Manager Performance & Condition Monitoring — May 2024 - Feb 2026

- Analyzed historical plant data and design metrics to build energy monitoring and root-cause analysis plan, reducing auxiliary power consumption from grid by ~10%.
- Authored data framework and SOP for annual 490 MW capacity tests, integrating complex performance correction formulas to ensure strict compliance with NEPRA regulatory standards.
- Monitored performance data for gas/steam turbines to calculate efficiencies, delivering daily internal analytics reports to eliminate operational inefficiencies
- Conducted a statistical impact study on seawater cooling temperatures, identifying root cause of seasonal degradation in steam turbine condensers and delivering data-driven mitigation strategies.

Lucky Renewables Pvt. Ltd. | *Data Analysis, Documentation* — Karachi, Pakistan

Management Trainee — Sept 2023 - May 2024

- Engineered daily and monthly Excel dashboards to track wind plant performance, delivering analytics to higher management that rapidly identified and corrected operational inefficiencies.
- Generated concise KPI reports and data-driven recommendations, collaborating with cross-functional engineering teams to implement plant improvements.
- Conducted thorough statistical data analysis to identify performance trends, detect operational anomalies, and isolate efficiency bottlenecks.

PROJECTS

Dog Breed Image Classification: Built a Python image-classification pipeline using pretrained CNNs (ResNet, AlexNet, and VGG) to identify dog breeds, distinguish dogs from non-dogs, and compare model accuracy against runtime. (Github Link)

Multi-Variate Data Correlation: Engineered strategic prompts within AWS PartyRock to visualize environmental datasets, identifying key correlations between carbon monoxide surges (3.8x baseline) and rush hour traffic patterns while validating AI-generated public health insights.

FitForge AI App: Developed generative AI application that automates repeatable planning workflows to dynamically generate personalized workout regimes, nutritional plans, and progress forecasts from user inputs. (App Link)

PROFESSIONAL CERTIFICATES

Future AWS AI Programmer (ongoing); CS50's Introduction to Databases with SQL by HarvardX (ongoing); Introduction to Data Analytics, IBM; AWS AI Practitioner Challenge, Udacity

EXTRA CURRICULAR ACTIVITIES

Management team Member for SPEC (Student Project Exhibition & Competition) and NED Literature Festival; Volunteered in Dar ul Sukun (Home for abandoned children with special needs).